WEEK 1 PROJECT REPORT

SUPERSTORE SALES DATA IMPORT, POWER QUERY & DATA CLEANING

Project Title: Superstore Sales Data Analysis using Power BI
Tool Used: Microsoft Power BI Desktop
Dataset: Superstore Sales Dataset

INTRODUCTION

During Week 1 of this project, I worked with the **Superstore Sales dataset** using **Microsoft Power BI Desktop**. The main objective was to understand the process of importing raw data and preparing it for further analysis.

The dataset contains sales transaction information related to customers, products, categories, regions, sales, discounts, quantities, and profits. Since raw datasets may contain inconsistent formatting, incorrect data types, duplicate records, blank values, or unstructured information, data preprocessing was performed using the **Power Query Editor**.

The major activities completed during this week were:

- Importing the Superstore dataset into Power BI
- Inspecting the dataset in Power Query
- Checking and correcting data types
- Identifying missing values and duplicate records
- Cleaning and standardizing text fields
- Transforming categorical attributes
- Creating useful date-based columns
- Applying the transformations to the Power BI data model
- Verifying the cleaned dataset

The cleaned dataset prepared during this stage will serve as the foundation for visualization and business analysis in the upcoming weeks.

OBJECTIVES

The main objectives of Week 1 were:

1. To understand how to import a raw dataset into Power BI.
2. To explore the structure of the Superstore dataset.
3. To learn how to use the Power Query Editor.
4. To identify and handle data quality issues.
5. To ensure appropriate data types for different columns.

6. To clean and standardize categorical data.
7. To create useful date and time attributes.
8. To load the transformed data into Power BI.
9. To prepare a reliable dataset for future analysis and dashboard development.

STEP 1 – IMPORTING DATA INTO POWER BI

The first step was to import the Superstore dataset into Microsoft Power BI Desktop.

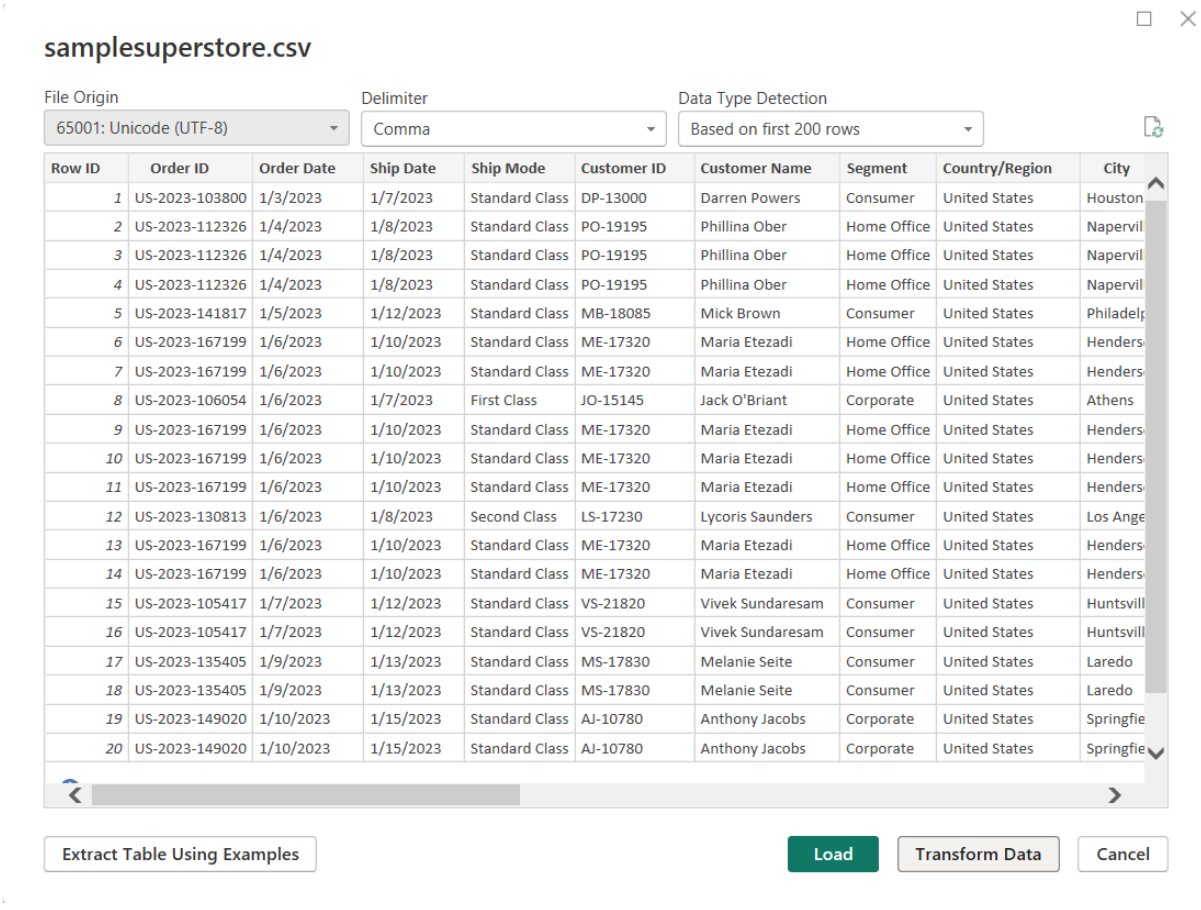
Home → Get Data → Text/CSV

Then, selected **Transform Data** so that the data could be inspected and cleaned using Power Query.

Power BI provides three major views:

- **Report View** – Used for creating charts, graphs, and dashboards.
- **Table View** – Used to inspect the actual rows and columns.
- **Model View** – Used to view tables and relationships.

This initial step helped me understand the structure and contents of the dataset before beginning the cleaning process.



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City
1	US-2023-103800	1/3/2023	1/7/2023	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston
2	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Napervil
3	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Napervil
4	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Napervil
5	US-2023-141817	1/5/2023	1/12/2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelp
6	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henders
7	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henders
8	US-2023-106054	1/6/2023	1/7/2023	First Class	JO-15145	Jack O'Briant	Corporate	United States	Athens
9	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henders
10	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henders
11	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henders
12	US-2023-130813	1/6/2023	1/8/2023	Second Class	LS-17230	Lycoris Saunders	Consumer	United States	Los Ange
13	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henders
14	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henders
15	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsvill
16	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsvill
17	US-2023-135405	1/9/2023	1/13/2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States	Laredo
18	US-2023-135405	1/9/2023	1/13/2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States	Laredo
19	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfie
20	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfie

STEP 2 – INSPECTING DATA IN POWER QUERY

After importing the dataset, I opened the **Power Query Editor** by selecting **Transform Data**. Power Query provides various tools for examining and transforming raw data. I inspected the columns and reviewed the information contained in each field.

The following aspects were checked:

- Column names
- Data types
- Missing or blank values
- Errors
- Duplicate records
- Text formatting
- Numerical values
- Date fields

I also enabled data-quality information where required to identify valid, empty, and error values.

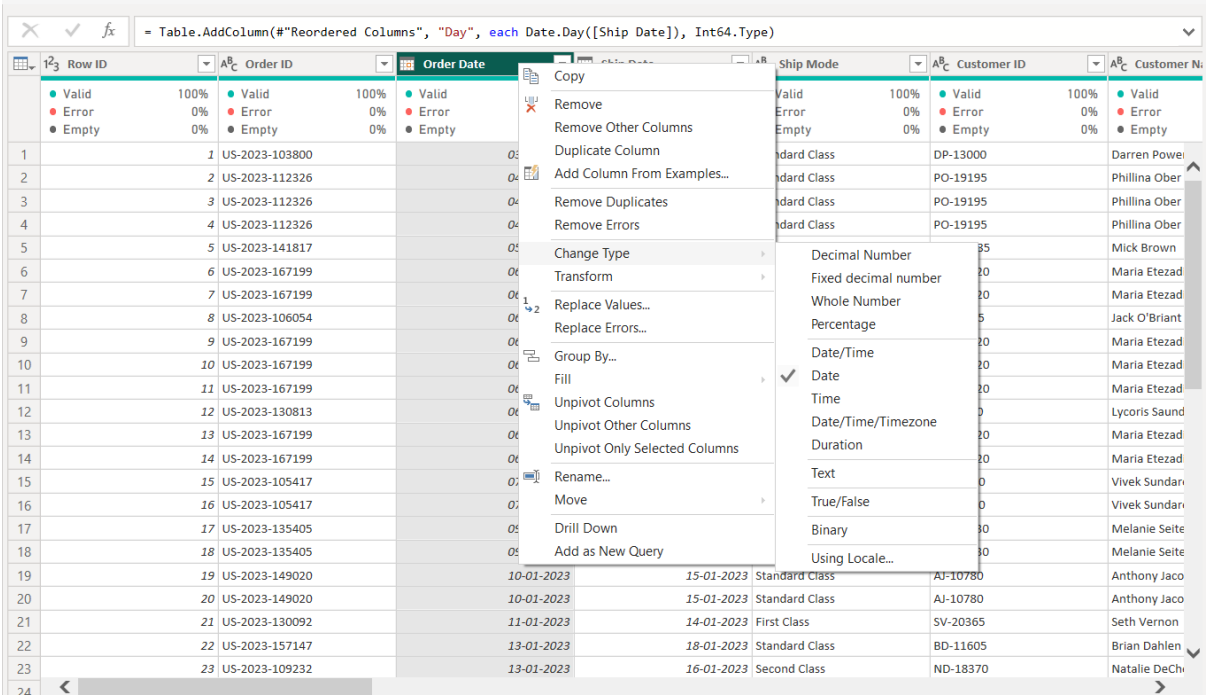
This inspection helped determine which transformations were required before the dataset could be used for analysis.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province
12	US-2023-130813	06 January 2023	08 January 2023	SECOND CLASS	LS-17230	LYCORIS SAUNDERS	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
57	US-2023-148614	20 January 2023	25 January 2023	STANDARD CLASS	MV-17485	MARK VAN HUFF	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
122	US-2023-164903	20 February 2023	24 February 2023	STANDARD CLASS	SR-20740	STEVEN ROELLE	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
331	US-2023-102652	06 April 2023	12 April 2023	STANDARD CLASS	AY-10555	ANDY YOTOV	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
332	US-2023-102652	06 April 2023	12 April 2023	STANDARD CLASS	AY-10555	ANDY YOTOV	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
370	US-2023-120740	15 April 2023	15 April 2023	SAME DAY	PS-18970	PAUL STEVENSON	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
373	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	JL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
375	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	JL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
377	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	JL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
723	US-2023-114125	09 July 2023	13 July 2023	STANDARD CLASS	GH-14410	GARY HANSEN	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
727	US-2023-114125	09 July 2023	13 July 2023	STANDARD CLASS	GH-14410	GARY HANSEN	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
1409	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1410	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1411	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1422	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1427	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1428	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1430	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1431	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1446	US-2023-168305	01 November 2023	08 November 2023	STANDARD CLASS	PL-18925	PAUL LUCAS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
1449	US-2023-168305	01 November 2023	08 November 2023	STANDARD CLASS	PL-18925	PAUL LUCAS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
1468	US-2023-153619	03 November 2023	03 November 2023	SAME DAY	TC-21535	TRACY COLLINS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
1816	US-2023-138737	07 December 2023	10 December 2023	FIRST CLASS	FP-14320	FRANK PREIS	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
1859	US-2023-113579	13 December 2023	15 December 2023	SECOND CLASS	KD-16345	KATHERINE DUCICH	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
2288	US-2024-138219	23 March 2024	30 March 2024	STANDARD CLASS	BP-11095	BART PISTOLE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
3056	US-2024-121188	28 August 2024	04 September 2024	STANDARD CLASS	CB-12025	CASSANDRA BRANDOW	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA

STEP 3 – CHECKING AND CORRECTING DATA TYPES

Correct data types are essential for accurate analysis.

For example, a date column should be recognized as a **Date**, while numerical fields such as Sales and Profit should be recognized as numerical values. Incorrect data types can lead to problems when performing calculations, filtering data or creating visualizations. Therefore, the data types were verified and corrected wherever necessary.



STEP 4 – CHECKING MISSING VALUES AND DUPLICATES

The dataset was examined for blank values and duplicate records.

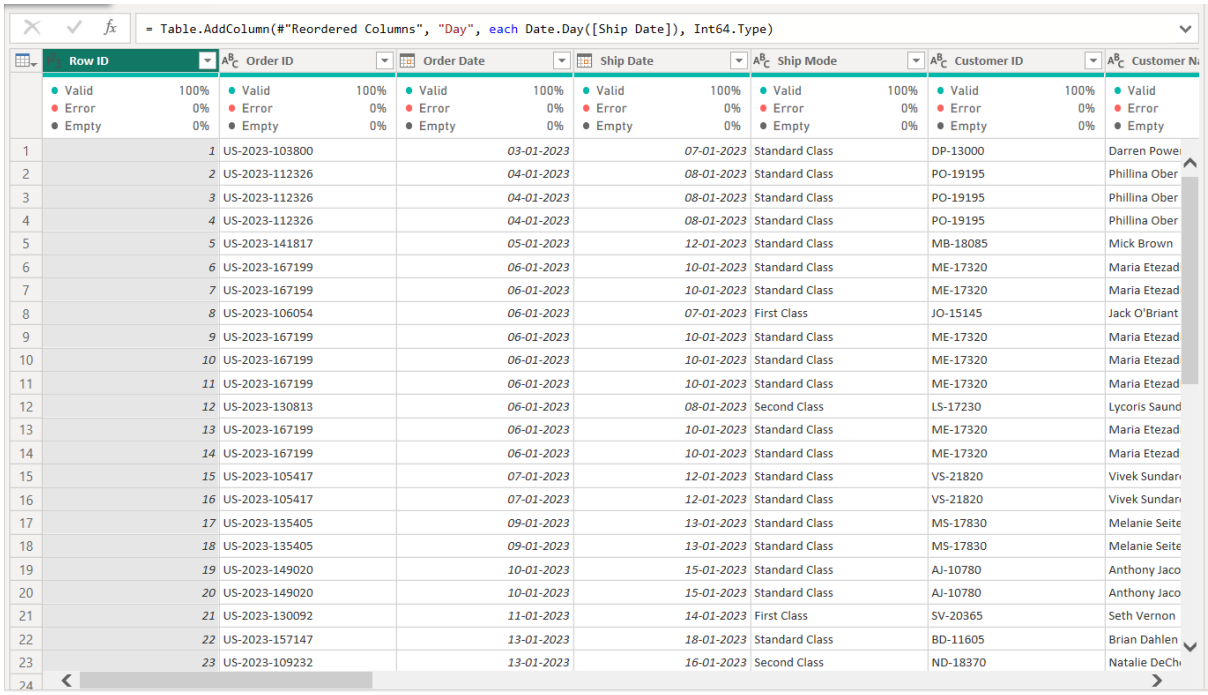
Power Query's data-quality features were used to identify empty and error values. Important fields were checked carefully because missing information in important columns can affect the accuracy of analysis.

Duplicate records were also checked to ensure that the same transaction was not unintentionally repeated.

Where appropriate, duplicate records can be removed using:

Home → Remove Rows → Remove Duplicates

The purpose of this step was to improve data reliability and prevent duplicate transactions from affecting future calculations.



= Table.AddColumn(#"Reordered Columns", "Day", each Date.Day([Ship Date]), Int64.Type)

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Na
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	1	US-2023-103800	03-01-2023	07-01-2023	Standard Class	DP-13000	Darren Powe
2	2	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
3	3	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
4	4	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
5	5	US-2023-141817	05-01-2023	12-01-2023	Standard Class	MB-18085	Mick Brown
6	6	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
7	7	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
8	8	US-2023-106054	06-01-2023	07-01-2023	First Class	JO-15145	Jack O'Briant
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17	17	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Seite
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19	19	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
20	20	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
21	21	US-2023-130092	11-01-2023	14-01-2023	First Class	SV-20365	Seth Vernon
22	22	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlen
23	23	US-2023-109232	13-01-2023	16-01-2023	Second Class	ND-18370	Natalie DeCh

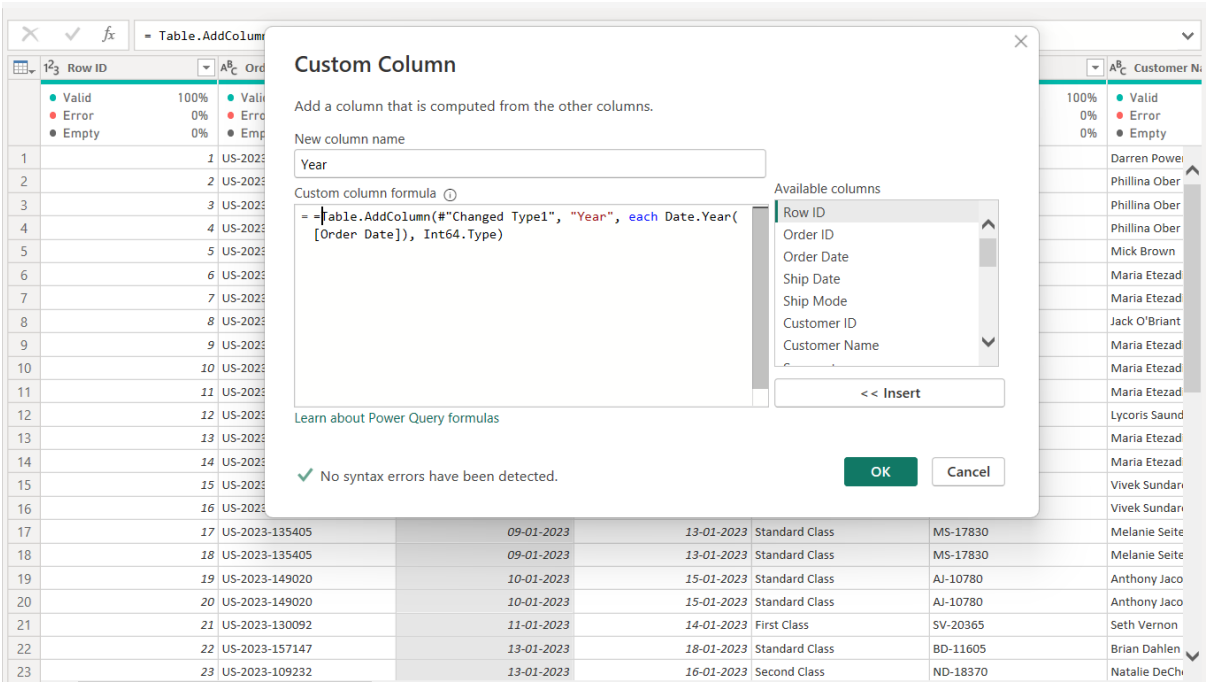
STEP 5 – CREATING YEAR COLUMN

The **Year** column was extracted from the Order Date column.

In Power Query, this was performed using:

Add Column → Date → Year → Year

The resulting column contains the year associated with each transaction.



STEP 6 – APPLYING THE TRANSFORMATIONS

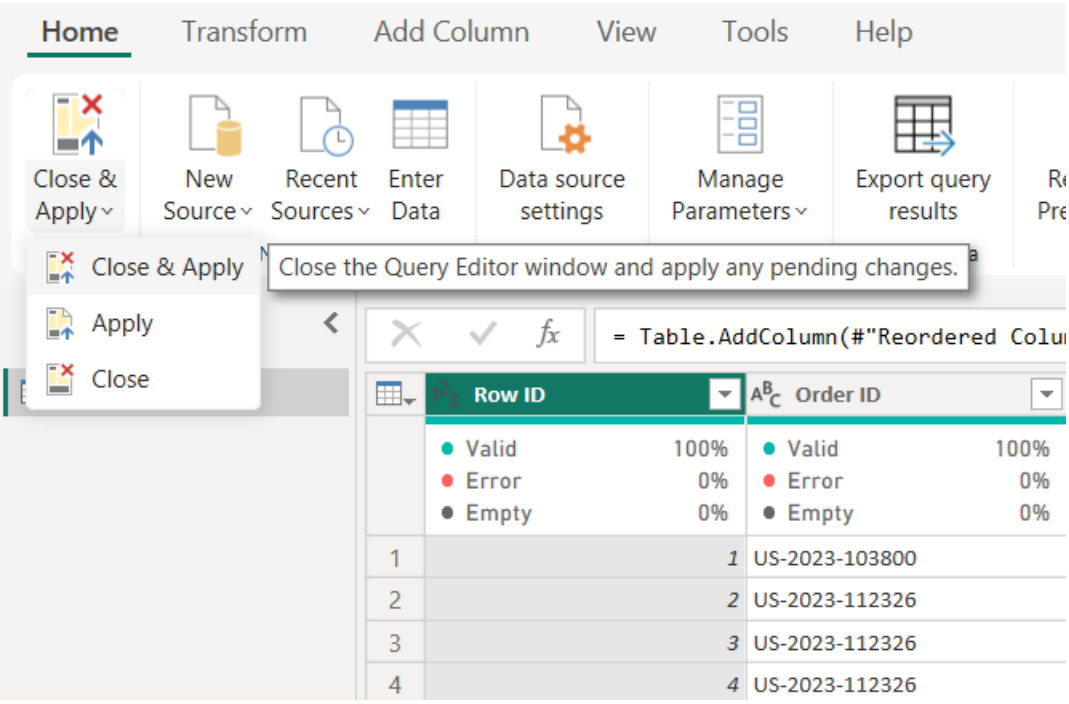
After completing the required cleaning and transformations, the final dataset was reviewed in Power Query.

The following aspects were verified:

- Data types were correct.
- Text fields were standardized.
- Important categorical attributes were consistent.
- Date attributes were created.
- Unwanted errors were checked.
- Duplicate records were reviewed.

Once the dataset was ready, I selected:

Home → Close & Apply



Power BI then loaded the transformed dataset into the main Power BI environment.

STEP 7 – VERIFYING THE CLEAN DATA

After applying the transformations, I opened **Data View** in Power BI Desktop.

The newly created date columns were checked to confirm that they had been successfully added.

The final dataset was now structured and ready for further analysis.

```
= Table.AddColumn(#"Reordered Columns", "Day", each Date.Day([Ship Date]), Int64.Type)
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Na
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	1	US-2023-103800	03-01-2023	07-01-2023	Standard Class	DP-13000	Darren Power
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23	23	US-2023-109232	13-01-2023	16-01-2023	Second Class	ND-18370	Natalie DeCh

DATA PREPROCESSING WORKFLOW

The complete Week 1 preprocessing workflow can be summarized as follows:

Raw Superstore Dataset

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Import into Power BI

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Open Power Query

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Inspect Data

↓

Check Data Types

↓

Check Missing Values & Duplicates

↓

Clean Text Fields

↓

Standardize Categories

↓

Create Date Attributes

↓

Verify Transformations

↓

Close & Apply

↓

Final Clean Dataset

↓

Ready for Analysis

KEY LEARNING OUTCOMES

Through this week's activity, I gained practical knowledge of the data preprocessing workflow in Power BI.

The major learning outcomes were:

- Understanding how to import external datasets into Power BI.
- Understanding the purpose of Power Query.
- Learning how to inspect raw data.
- Learning how to change column data types.
- Understanding the importance of data quality.
- Learning how to clean text values.
- Understanding categorical data standardization.
- Learning how to extract date-based attributes.
- Understanding basic Power Query M functions.
- Learning how to apply transformations and load the final dataset.
- Understanding how clean data provides a foundation for reliable business intelligence.

CONCLUSION

In Week 1, I successfully completed the initial data preparation stage of the Superstore Sales Analysis project using Microsoft Power BI.

The raw dataset was imported into Power BI and examined using Power Query Editor. Data types were verified, data quality was checked, text fields were cleaned, categorical attributes were standardized, and useful date-based columns were created.

The **Close & Apply** operation was then used to load the transformed dataset into Power BI. The final dataset was verified in Data View and Model View to ensure that the required transformations were successfully applied.

This activity demonstrated that **data preprocessing is a critical step in business intelligence and data analytics**. A properly cleaned and structured dataset improves the reliability of calculations, visualizations, and business insights.

The cleaned Superstore dataset is now ready for the next stages of the project, where it can be used to create visualizations, analyze sales and profit trends, identify customer and product patterns, and develop an interactive Power BI dashboard.